complex, far more so than quantum mechanics. Einstein famously had to enlist the help of a mathematician friend, Marcel Grossman, just to learn the differential geometry necessary to formulate and understand his own theory. This combination of unfamiliar subject matter and obscure mathematics made it difficult to be sure of what the theory was saying and led many physicists to be suspicious of its conclusions. Even Einstein himself had trouble accepting the consequences of his own theory after he initially developed and published it in 1915. He realized that general relativity implied the universe as a whole must be either contracting or expanding, a conclusion he found troubling, at odds with all known data at the time. So he put in a "cosmological constant," a fudge factor to keep the universe at a static size. But, in 1929, the astronomer Edwin Hubble discovered that distant galaxies appeared to be receding at a rate proportional to their distance—exactly what you would expect to see in an expanding universe. Einstein readily dismissed his ad hoc cosmological constant—he had never liked it anyhow, calling it "gravely detrimental to the beauty of the theory"—and accepted the cosmological picture presented by general relativity as correct. But not everyone was convinced—including Edwin Hubble himself. Hubble and others thought that distant galaxies only appeared to be receding and that the universe was in fact static. Others agreed the universe was expanding, but they proposed modifications to the laws of physics that would ensure the universe looked basically the same at all times in the past and future, despite its expansion. This came to be known as the "steady-state" theory of the universe. For decades afterward, the steady-state theory was considered a reasonable scientific theory, and many physicists found it quite a bit more reasonable than the expanding universe of general relativity. After all, such a universe must have once been in a fabulously hot, dense, and small state, expanding rapidly—what steady-state theorist Fred Hoyle termed a "Big Bang." General relativity, Hoyle and others held, was not necessarily to be trusted under such strange conditions, nor when applied to the universe as a whole.

Meanwhile, confusion persisted about the implications of general relativity even for objects much smaller than the entire universe, like stars. In 1938, Robert Oppenheimer and his student George Volkoff at Berkeley, along with Richard Tolman at Caltech, used a very early forerunner of a computer to calculate that supermassive stars, far larger than our Sun, must